Lab 4 Activity

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

1

Load in the `CO2` data from the `datasets` package. Look at the documentation to understand the variables in this dataset. Identify a way to analyze these data using a *t* test (several possible choices, but `conc` probably won't be an interesting DV). Evaluate each assumption, conduct the *t* test, compute an effect size, and interpret the results.

```r
data(CO2, package = "datasets")
?CO2
CO2 %>% summary
```

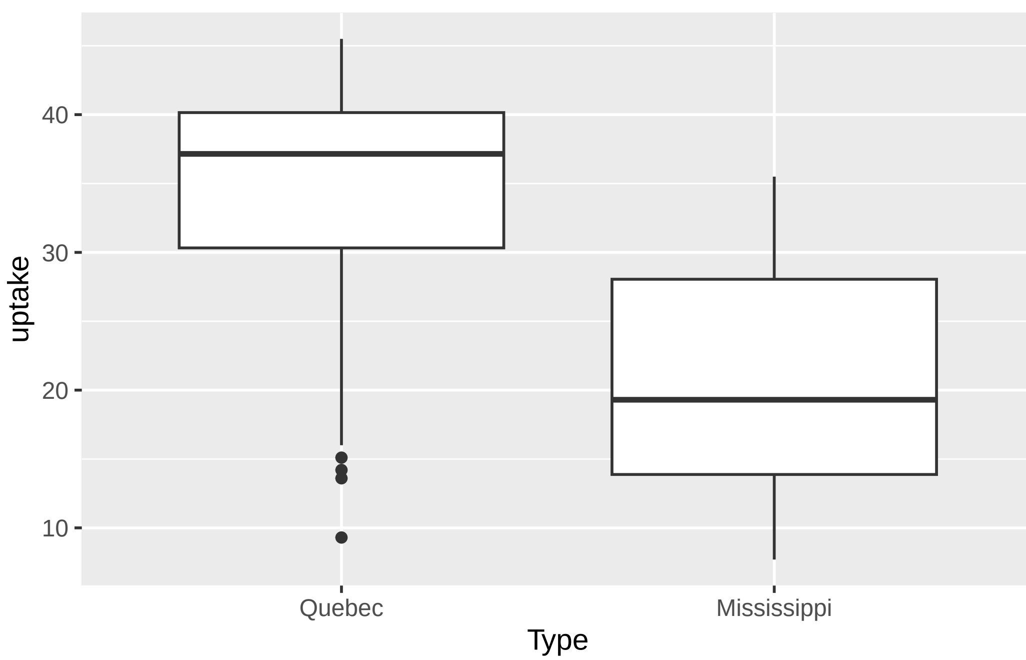
```
     Plant             Type         Treatment       conc          uptake     
 Qn1    : 7   Quebec     :42   nonchilled:42   Min.   :  95   Min.   : 7.70  
 Qn2    : 7   Mississippi:42   chilled   :42   1st Qu.: 175   1st Qu.:17.90  
 Qn3    : 7                                    Median : 350   Median :28.30  
 Qc1    : 7                                    Mean   : 435   Mean   :27.21  
 Qc3    : 7                                    3rd Qu.: 675   3rd Qu.:37.12  
```

```
 Qc2     : 7                                         Max.   :1000    Max.   :45.50
 (Other):42
```

```
# Type or Treatment are appropriate IVs for an ind. samples t-test
# uptake is the most obvious DV

## Type as the IV:

# test for outliers:
CO2 %>% group_by(Type) %>%
  summarize(mean = mean(uptake),
            tmean = mean(uptake, trim = .05),
            .groups = "keep") # some differences for Quebec
```

```
# A tibble: 2 x 3
# Groups:   Type [2]
  Type         mean tmean
  <fct>       <dbl> <dbl>
1 Quebec       33.5  34.1
2 Mississippi  20.9  20.8
```

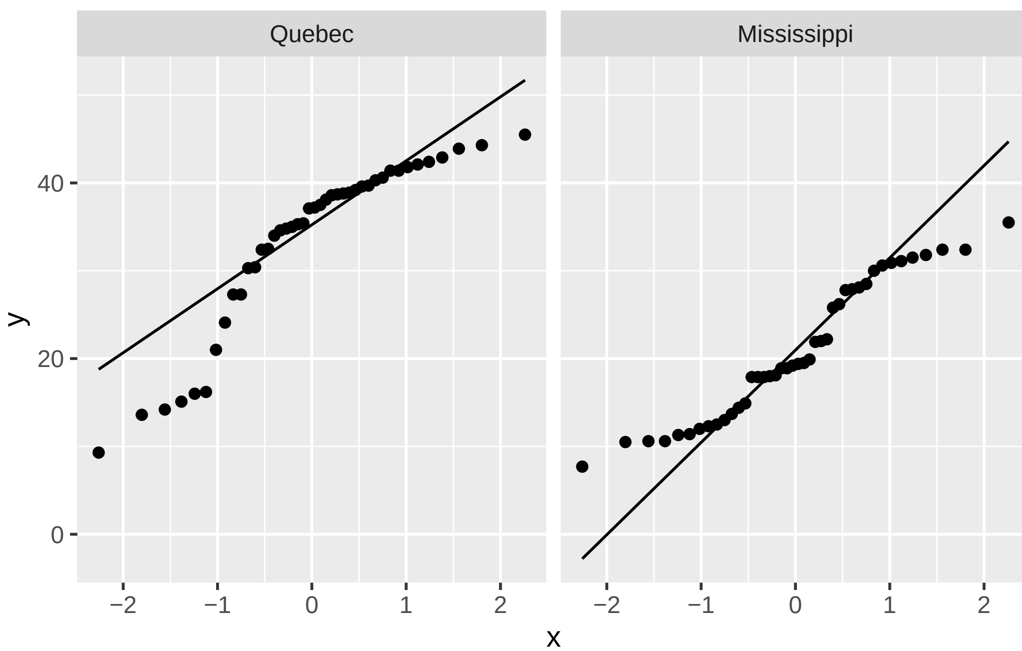
```
ggplot(CO2, aes(Type, uptake)) + geom_boxplot()
```



```
# possibly some outliers for Quebec

# test for normality:
library(e1071)
CO2 %>% group_by(Type) %>%
  summarize(mean = mean(uptake), sd = sd(uptake),
            skew = skewness(uptake), kurt = kurtosis(uptake),
            n = n(), .groups = "keep")
```

```
# A tibble: 2 x 6
# Groups:   Type [2]
  Type         mean    sd   skew    kurt     n
  <fct>       <dbl> <dbl>  <dbl>   <dbl> <int>
1 Quebec       33.5  9.67 -1.05  -0.0922    42
2 Mississippi  20.9  7.82  0.172 -1.32      42
```

```
# some potentially meaningful skew for Quebec, kurt for Mississippi
ggplot(CO2, aes(sample = uptake)) + stat_qq() +
  stat_qq_line() + facet_wrap(~Type) # some major violations of normality
```



```r
# but 42/group might mitigate against this violation

# test for homogeneity of variance:
library(car)
```

```
Loading required package: carData


Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```r
leveneTest(uptake ~ Type, CO2)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.1704 0.6808
      82
```

```r
# homogeneity not violated, but we'll use Welch anyways

# t-test (two-tailed, Welch)
t.test(uptake ~ Type, CO2)
```

```

    Welch Two Sample t-test

data:  uptake by Type
t = 6.5969, df = 78.533, p-value = 4.451e-09
alternative hypothesis: true difference in means between group Quebec and group Mississippi i
95 percent confidence interval:
  8.839475 16.479572
sample estimates:
     mean in group Quebec mean in group Mississippi
                 33.54286                  20.88333
```

```r
# effect size - put pooled standard deviation in the denominator
CO2 %>% group_by(Type) %>% summarize(var(uptake), n(), .groups = "keep")
```

```
# A tibble: 2 x 3
# Groups:   Type [2]
  Type        `var(uptake)` `n()`
  <fct>               <dbl> <int>
1 Quebec               93.6    42
2 Mississippi          61.1    42
```

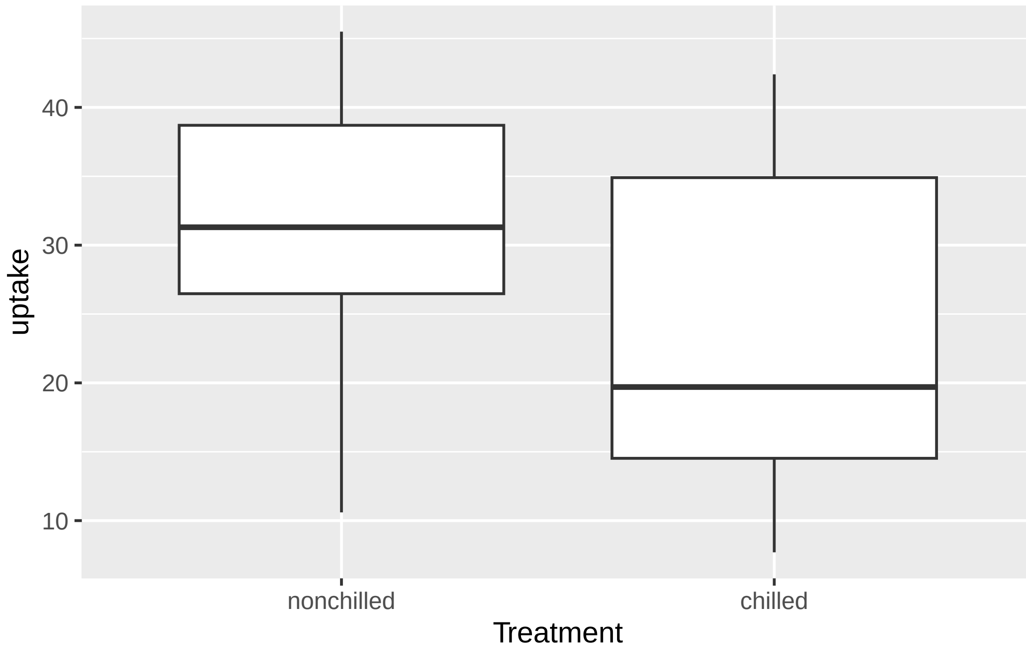
```r
sp <- sqrt((41 * 93.6 + 41 * 61.1) / (42 + 42 - 2))
(33.54286 - 20.88333) / sp
```

```
[1] 1.43942
```

```r
# mean CO2 uptake rate in Quebec is significantly higher than in Mississippi with a very larg


## Treatment as the IV:

# test for outliers:
CO2 %>% group_by(Treatment) %>%
  summarize(mean = mean(uptake),
            tmean = mean(uptake, trim = .05),
            .groups = "keep") # no major differences
```

```
# A tibble: 2 x 3
# Groups:   Treatment [2]
  Treatment   mean tmean
  <fct>      <dbl> <dbl>
1 nonchilled  30.6  30.9
2 chilled     23.8  23.6
```

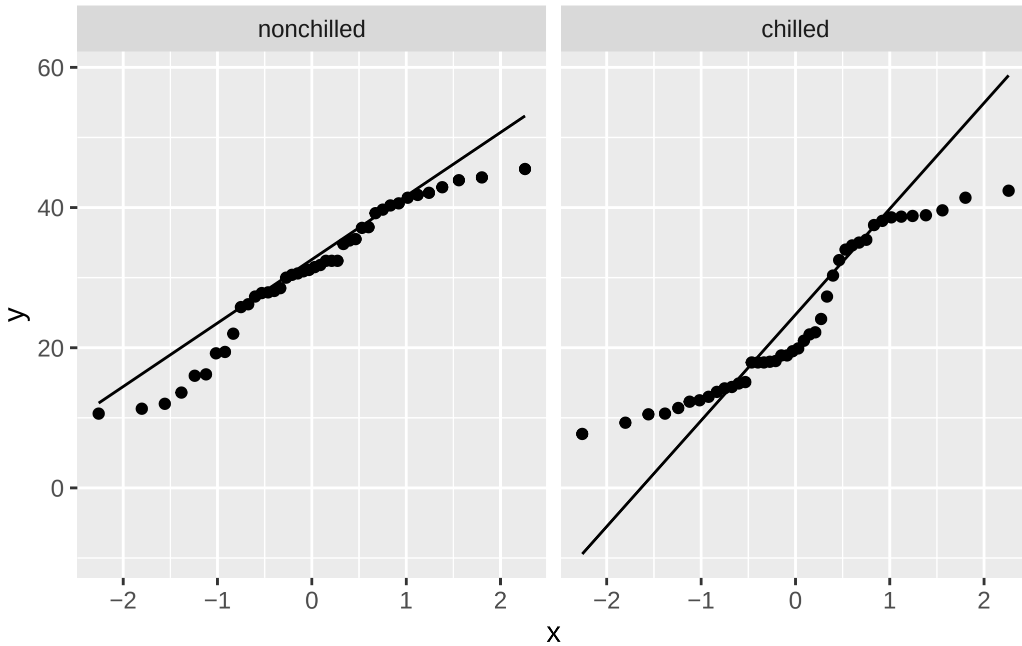
```r
ggplot(CO2, aes(Treatment, uptake)) + geom_boxplot()
```




```
# no outliers shown

# test for normality:
library(e1071)
CO2 %>% group_by(Treatment) %>%
  summarize(mean = mean(uptake), sd = sd(uptake),
            skew = skewness(uptake), kurt = kurtosis(uptake),
            n = n(), .groups = "keep")
```

```
# A tibble: 2 x 6
# Groups:   Treatment [2]
  Treatment   mean    sd   skew   kurt     n
  <fct>      <dbl> <dbl>  <dbl>  <dbl> <int>
1 nonchilled  30.6  9.70 -0.482 -0.706    42
2 chilled     23.8 10.9   0.327 -1.45     42
```

```
# some potentially meaningful skew and kurt for both groups
ggplot(CO2, aes(sample = uptake)) + stat_qq() +
  stat_qq_line() + facet_wrap(~Treatment) # some major violations of normality
```




```
# but 42/group might mitigate against this violation

# test for homogeneity of variance:
library(car)
leveneTest(uptake ~ Treatment, CO2)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  1.2999 0.2576
      82
```

```
# homogeneity not violated, but we'll use Welch anyways

# t-test (two-tailed, Welch)
t.test(uptake ~ Treatment, CO2)
```

```

    Welch Two Sample t-test

data:  uptake by Treatment
t = 3.0485, df = 80.945, p-value = 0.003107
```

```
alternative hypothesis: true difference in means between group nonchilled and group chilled i
95 percent confidence interval:
  2.382366 11.336682
sample estimates:
mean in group nonchilled    mean in group chilled
                30.64286                 23.78333
```

```
# effect size - put pooled standard deviation in the denominator
CO2 %>% group_by(Treatment) %>% summarize(var(uptake), n(), .groups = "keep")
```

```
# A tibble: 2 x 3
# Groups:   Treatment [2]
  Treatment  `var(uptake)` `n()`
  <fct>              <dbl> <int>
1 nonchilled          94.2    42
2 chilled            118.     42
```

```
sp <- sqrt((41 * 94.2 + 41 * 118) / (42 + 42 - 2))
(30.64286 - 23.78333) / sp
```

```
[1] 0.6659424
```

```
# mean CO2 uptake rate for nonchilled treatment is significantly higher than in for the chil
```

2

Load in the `sleep` data from the `datasets` package. Look at the documentation to understand the variables in this dataset (but try not to peek at the examples section of the documentation). Identify a way to analyze these data using a t test. Evaluate each assumption, conduct the t test, compute an effect size, and interpret the results.

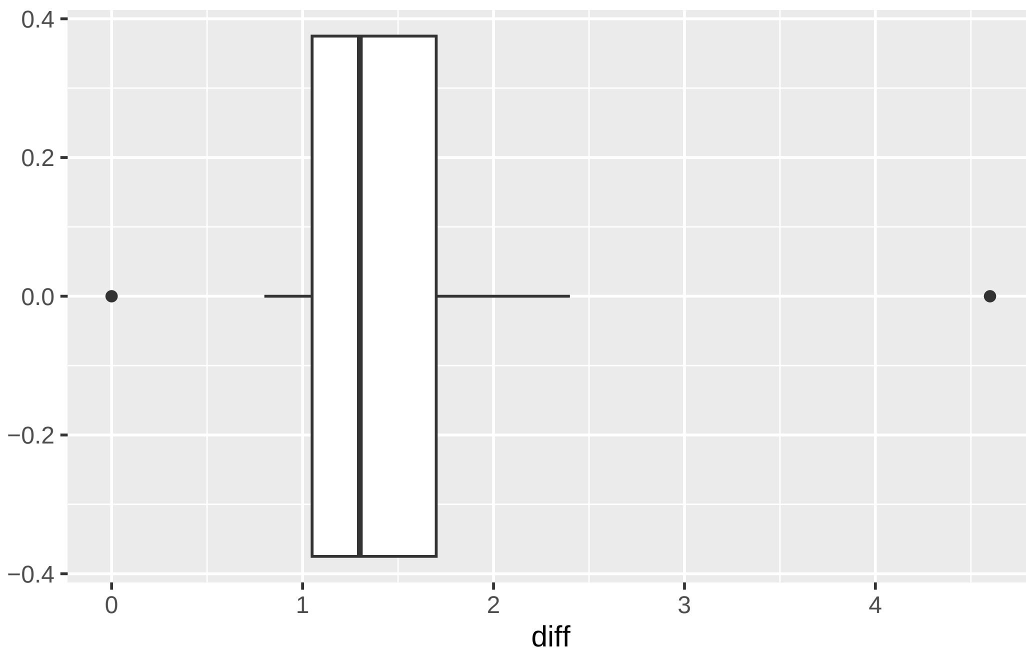
Hint: run the following code to transform the data into "wide" format, which will make the dataset easier to work with:

```
sleep2 <- sleep %>%
  pivot_wider(id_cols = ID,
              names_from = group,
              names_prefix = "group",
              values_from = extra)
```

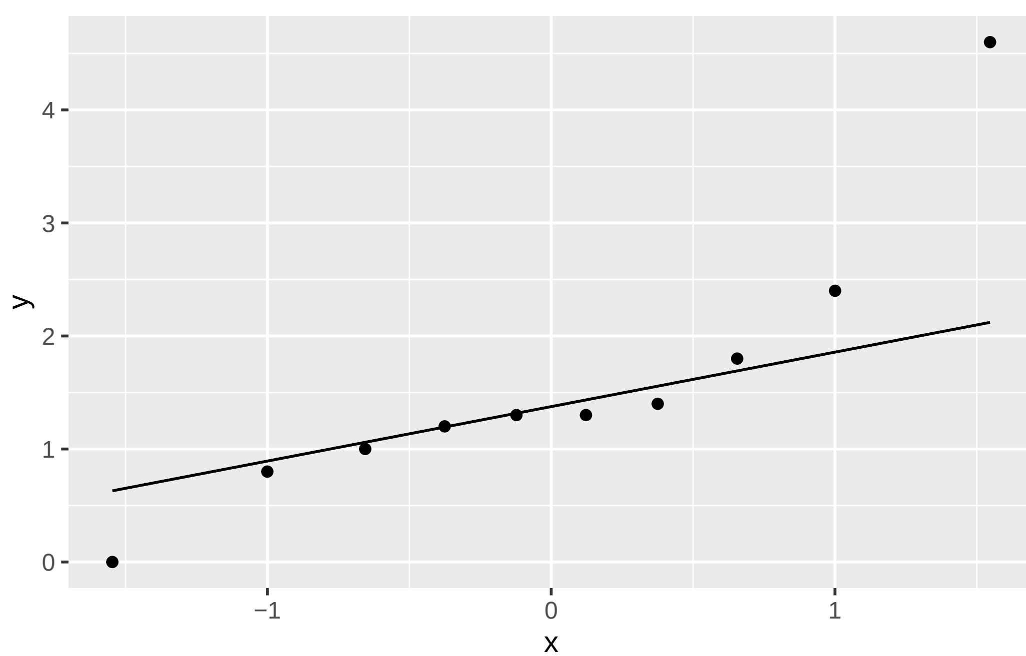
```r
data(sleep, package = "datasets")
sleep2 <- sleep %>%
  pivot_wider(id_cols = ID,
              names_from = group,
              names_prefix = "group",
              values_from = extra)
sleep2 %>% summary
```

```
       ID        group1           group2
 1      :1   Min.   :-1.600   Min.   :-0.100
 2      :1   1st Qu.:-0.175   1st Qu.: 0.875
 3      :1   Median : 0.350   Median : 1.750
 4      :1   Mean   : 0.750   Mean   : 2.330
 5      :1   3rd Qu.: 1.700   3rd Qu.: 4.150
 6      :1   Max.   : 3.700   Max.   : 5.500
 (Other):4
```

```r
# extra should be the DV
# group is an appropriate IV for paired samples t-test (see documentation)

# want to evaluate the normality of difference scores
sleep2 <- sleep2 %>% mutate(diff = group2 - group1)

# trimmed mean may not be appropriate because N is so small
ggplot(sleep2, aes(diff)) + geom_boxplot()
```




```
# possibly likely outliers, but hard to draw any definitive conclusions

ggplot(sleep2, aes(sample = diff)) + stat_qq() + stat_qq_line() # probably non-normal
```



```r
# an upper-tailed test may be appropriate if we are only concerned about gains
t.test(sleep2$diff, alternative = "greater")
```

```
    One Sample t-test

data:  sleep2$diff
t = 4.0621, df = 9, p-value = 0.001416
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.8669947       Inf
sample estimates:
mean of x
     1.58
```

```r
# effect size
4.0621 / sqrt(10)
```

```
[1] 1.284549
```

```r
# there is a significant increase in sleep with drug 2 over drug 1 with a large effect size
```